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Torque converter with pulling bias spring

Abstract

A torque converter includes: a front cover arranged to receive torque; an impeller having an impeller shell non-rotatably connected to the front cover; a turbine in fluid communication with the impeller and including a turbine shell; and a damper assembly disposed axially between the front cover and the turbine shell. The damper assembly includes a first cover plate non-rotatably connected to the turbine shell. The damper assembly further includes a second cover plate non-rotatably connected to the first cover plate and disposed axially between the front cover and the first cover plate. The damper assembly further includes an intermediate flange disposed axially between the first cover plate and the second cover plate. The damper assembly further includes a bias spring engaged with the intermediate flange and the second cover plate. The bias spring is configured to pull the second cover plate into contact with the intermediate flange.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to torque converters. More specifically, embodiments of the disclosure relate to a diaphragm spring for a torque converter.

BACKGROUND

(2) Many vehicles include a launch device between the engine and the transmission. A torque converter is a type of launch device commonly used in vehicles having an automatic transmission. A typical torque converter includes an impeller fixed to the crankshaft of the engine and a turbine fixed to a turbine shaft, which is the input to the transmission. To improve fuel economy, many torque converters include a bypass or lockup clutch that mechanically couples the turbine to a case of the torque converter to bypass the fluid coupling.

(3) In some embodiments, a damper assembly for a torque converter may include a bias spring used to push other damper components into contact with each other to create friction and thereby improve NVH (i.e., “noise,” “vibration,” and “harshness”) performance of the damper assembly. However, such a bias spring can increase a size of the damper assembly due to a stack path required to utilize the bias spring to create the contact between the intermediate flange and the one cover plate. Due to limited spacing within a torque converter envelope, it is desirable to have alternative designs and configurations to fit all the necessary components within the torque converter while still meeting durability and performance requirements.

SUMMARY

(4) Embodiments of the present disclosure provide a torque converter having a front cover arranged to receive torque. The torque converter further includes an impeller having an impeller shell non-rotatably connected to the front cover. The torque converter further includes a turbine in fluid communication with the impeller and including a turbine shell. The torque converter further includes a damper assembly disposed axially between the front cover and the turbine shell. The damper assembly includes a first cover plate non-rotatably connected to the turbine shell. The damper assembly further includes a second cover plate non-rotatably connected to the first cover plate and disposed axially between the front cover and the first cover plate. The damper assembly further includes an intermediate flange disposed axially between the first cover plate and the second cover plate. The damper assembly further includes a bias spring engaged with the intermediate flange and the second cover plate. The bias spring is configured to pull the second cover plate into contact with the intermediate flange.

(5) In embodiments, the damper assembly may further include a drive flange disposed axially between the first and second cover plates. The drive flange may be arranged to non-rotatably connect to a transmission input shaft. The intermediate flange may be disposed axially between the drive flange and the second cover plate. The intermediate flange may be radially spaced from the transmission input shaft. In embodiments, the torque converter may further include a further bias spring engaged with the turbine shell and the first cover plate. The further bias spring may be configured to push the first cover plate into contact with the drive flange. The further bias spring may be disposed radially inside of the intermediate flange.

(6) In embodiments, the damper assembly may further include an arc spring supported by the first cover plate and the second cover plate. The bias spring may be disposed radially inside of the arc spring. The intermediate flange and the second cover plate may be configured to contact each other radially outside of the arc spring.

(7) In embodiments, the intermediate flange may include an axially extending portion disposed radially inside of the second cover plate. The bias spring may be engaged with the axially extending portion. The axially extending portion may include a slot extending, at least partially, radially therethrough. The bias spring may be disposed, at least partially, in the slot. The axially extending portion may include a flange extending radially outwardly from the axially extending portion. The bias spring may be in contact with the flange. The axially extending portion may include two radial stops circumferentially spaced from each other and configured to radially constrain the bias spring. The flange may be disposed circumferentially between the two radial stops.

(8) In embodiments, the bias spring may include a body in contact with the second cover plate and a tab extending radially inwardly from the body. The body may be disposed axially between the second cover plate and the front cover. The intermediate flange may include a retaining feature disposed axially between the body and the first cover plate. The retaining feature may be configured to deflect the tab relative to the body towards the first cover plate.

(9) Embodiments of the present disclosure further provide a damper assembly for a torque converter. The damper assembly includes a first cover plate arranged to receive torque. The damper assembly further includes a second cover plate non-rotatably connected to the first cover plate and axially spaced from the first cover plate. The damper assembly further includes an intermediate flange disposed axially between the first cover plate and the second cover plate. The damper assembly further includes a bias spring engaged with the intermediate flange and the second cover plate. The bias spring is configured to pull the second cover plate into contact with the intermediate flange.

(10) In embodiments, the damper assembly may further include an arc spring supported by the first cover plate and the second cover plate. The bias spring may be disposed radially inside of the arc spring. The intermediate flange and the second cover plate may be configured to contact each other

radially outside of the arc spring.
(11) In embodiments, the bias spring may include a body in contact with the second cover plate and a tab extending radially inwardly from the body. The second cover plate may be disposed, at least partially, between the body and the first cover plate. The intermediate flange may include a retaining feature disposed axially between the body and the first cover plate. The retaining feature may be configured to deflect the tab relative to the body towards the first cover plate.
(12) In embodiments, the intermediate flange may include an axially extending portion disposed radially inside of the second cover plate. The bias spring may be engaged with the axially extending portion.
(13) In embodiments, the axially extending portion may include a retaining feature configured to engage the bias spring. The retaining feature may be one of a slot or a radially extending flange.
(14) Embodiments presented herein provide the advantageous benefit of providing a bias spring configured to pull damper assembly components into contact with each other, which reduces a stack path for creating friction between the components. Further, embodiments disclosed herein offer design advantages by reducing a variation in install height of the pulling bias spring, which can result in more consistent hysteresis, loads, and stresses for the pulling bias spring while satisfying packaging constraints of the torque converter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates a cross-sectional side view of a torque converter including one embodiment of an intermediate flange according to the present disclosure; and
(2) FIG. **2** illustrates an enlarged view of an area of the torque converter shown in FIG. **1**;
(3) FIG. **3** a cross-sectional side view of a torque converter including an alternative embodiment of the intermediate flange according to the present disclosure;
(4) FIG. **4** illustrates an enlarged view of the area of the torque converter shown in FIG. **3**;
(5) FIG. **5** illustrates an enlarged view of the area of the torque converter shown in FIG. **3** according to an alternative embodiment of the intermediate flange;
(6) FIG. **6** illustrates a perspective view of a portion of an intermediate flange according to the embodiment shown in FIG. **5**.
DETAILED DESCRIPTION
(7) Embodiments of the present disclosure are described herein. It should be appreciated that like drawing numbers appearing in different drawing views identify identical, or functionally similar, structural elements. Also, it is to be understood that the disclosed embodiments are merely examples and other embodiments can take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the embodiments. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures can be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.
(8) The terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the present disclosure. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary

skill in the art to which this disclosure belongs. Although any methods, devices or materials similar or equivalent to those described herein can be used in the practice or testing of the disclosure, the following example methods, devices, and materials are now described.

(9) Referring to FIGS. **1-2**, a portion of a torque converter **100** is illustrated according to one embodiment of the present disclosure. At least some portions of the torque converter **100** are rotatable about a central axis A. While only a portion of the torque converter **100** above the central axis A is shown in FIG. **1**, it should be understood that the torque converter **100** can appear substantially similar below the central axis A with many components extending about the central axis A. Words such as “axial,” “radial,” “circumferential,” “outward,” etc. as used herein are intended to be with respect to the central axis A.

(10) The torque converter **100** includes: a front cover **102** arranged to receive torque; an impeller assembly **104**; a turbine assembly **106**; and a damper assembly **108**. The impeller assembly **104** includes: an impeller shell **110** non-rotatably connected to the front cover **102**; at least one impeller blade **112** attached to an inner surface of the impeller shell **110**; and an impeller hub **114** fixed to a radially inner end of the impeller shell **110**. The impeller shell **110** includes a blade supporting portion **116** for supporting the impeller blade(s) **112**. Radially outside of blade supporting portion **116**, a radially extending wall **118**, which forms an impeller clutch, radially protrudes outwardly from an outer circumference of rounded blade supporting portion **116**. Accordingly, the impeller clutch and impeller shell **110** are formed as a single piece. By “non-rotatably connected” components, we mean that: the components are connected so that whenever one of the components rotate, all the components rotate; and relative rotation between the components is not possible. Radial and/or axial movement of non-rotatably connected components with respect to each other is possible, but not required.

(11) The turbine assembly **106** is configured to define a piston that is axially moveable toward and away from the impeller assembly **104** to engage and disengage the impeller clutch so as to form a lock-up clutch. The turbine assembly **106** includes a turbine shell **120** and at least one turbine blade **122** attached to the turbine shell **120**. The turbine shell **120** includes a blade supporting portion **124** for supporting the turbine blade(s) **122**. Radially outside of the blade supporting portion **124**, the turbine shell **120** includes an outer radial extension **126**, which forms the piston and radially protrudes outwardly from an outer circumference of the blade supporting portion **124**. Accordingly, the piston and the turbine shell **120** are formed as a single piece. Radially inside of the blade supporting portion **124**, the turbine shell **120** includes an inner radial extension **128** that, at an inner radial end thereof, joins an axially extending inner circumferential section **130**, whose inner circumferential surface contacts an outer circumferential surface of a hub **132**.

(12) A friction material (not numbered) is bonded onto a surface of outer radial extension **126** for engaging radially extending wall **118**. In some embodiments, instead of or in addition to being bonded to the outer radial extension **126**, friction material may be bonded to radially extending wall **118**.

(13) The damper assembly **108**, which together with the turbine assembly **106** form a drive assembly of the torque converter **100**, is positioned between the front cover **102** and the turbine assembly **106** and is configured for transferring torque from the turbine assembly **106** to a transmission input shaft (not shown). The damper assembly **108** includes two cover plates-a turbine side cover plate **136**, which is fixed to the inner radial extension **128**, e.g., via rivets, and a front cover side cover plate **138**, which is disposed axially between the front cover **102** and the turbine side cover plate **136**. The cover plates **136**, **138** support a set of arc springs **140** axially therebetween.

(14) The damper assembly **108** also includes a drive flange **142** positioned axially between the cover plates **136**, **138**. The drive flange **142** is configured for non-rotatably connecting to the transmission input shaft. Radially outside of the arc springs **140**, the cover plates **136**, **138** are fixed together by a plurality of circumferentially spaced connectors (not numbered), e.g., rivets. The

connectors pass through the cover plates **136**, **138** into circumferential spaces (not numbered) formed between outer tabs (not numbered) extending from a radial outer end of the drive flange **142**.

(15) The damper assembly **108** further includes an intermediate flange **144** having a body **146** disposed axially between the cover plates **136**, **138**, and specifically between the drive flange **142** and the front cover side cover plate **138**. The intermediate flange **144** and the drive flange **142** each engage with the arc springs **140** to transmit torque through the damper assembly **108**. The body **146** and the drive flange **142** each include circumferentially extending slots (not numbered) for receiving respective arc springs **140**. The slots in the body **146** may, for example, be circumferentially offset relative to the slots in the drive flange **142** such that one slot of the body **146**, at least partially, overlaps circumferentially adjacent slots of the drive flange **142**. The intermediate flange **144** and the drive flange **142** may be configured to rotate relative to each other to transmit torque through the damper assembly **108**.

(16) The intermediate flange **144** further includes an axially extending portion **148** arranged at an inner diameter of the body **146**. The axially extending portion **148** is arranged radially inside of an inner diameter of the front cover side cover plate **138**. The axially extending portion **148** extends from the body **146** to an end **150**. The end **150** is disposed between the front cover **102** and the body **146**. The axially extending portion **148** may, for example, extend entirely circumferentially, i.e., continuously, about the central axis A. As another example, the axially extending portion **148** may extend discontinuously about the central axis A. That is, the axially extending portion **148** may include circumferentially adjacent segments with gaps therebetween.

(17) The outer radial extension **126** of the turbine assembly **106** engages the impeller assembly **104** at the radially extending wall **118** via the friction material to transfer torque input into the front cover **102** by an engine crankshaft to the transmission input shaft. As the turbine assembly **106** is driven by the impeller assembly **104**, either through contact via the friction material and the impeller shell **110** when the lockup clutch is locked or through fluid flow between blades **114**, **120**, the turbine assembly **106** transfers torque to the turbine side cover plate **136**. The cover plates **136**, **138** and the intermediate flange **144** transfer torque from the turbine assembly **106** to the drive flange **142** via the arc springs **140**. The drive flange **142** in turn drives the transmission input shaft.

(18) The damper assembly **108** may further include a pushing bias spring **152**, which in this embodiment is a diaphragm spring, that is axially between the drive flange **142** and the turbine assembly **106**. The pushing bias spring **152** engages the turbine side cover plate **136** and the drive flange **142**. The pushing bias spring **152** is preloaded, i.e., compressed between the drive flange **142** and the turbine assembly **106**, during installation and does not relax to a free state. When in drive, there is no axial force being transmitted by the pushing bias spring **152**, and therefore the damper assembly **108** has no additional hysteresis, which may be beneficial for NVH (i.e., “noise,” “vibration,” and “harshness”) performance, and the performance during a shift event can be equal to that of a baseline torque converter. Additionally, the damper assembly **108** may be arranged and configured to limit the axial force generated by the pushing bias spring **152** to prevent the piston from self-locking during the coast condition. More specifically, the maximum load produced in the coast condition may be limited by the load characteristics of the pushing bias spring **152** to prevent the lockup clutch from self-locking at high coast torques during the coast condition.

(19) The damper assembly **108** may further include a centering plate **154**, fixed to the front cover **102**, e.g. via a projection weld, and a bushing **156** press-fitted into the centering plate **154**. That is, the bushing **156** must be forcefully installed into the centering plate **154** such that the two are fixed together. The hub **132** may be installed into the bushing **156** and rotatable relative to the bushing **156**.

(20) The damper assembly **108** further includes a pulling bias spring **158**, which in this embodiment is a diaphragm spring, that is axially between the end **150** of the axially extending portion **148** and the front cover side cover plate **138**. The pulling bias spring **158** engages the front

cover side cover plate **138** and the axially extending portion **148**. The pulling bias spring **158** is preloaded, i.e., deflected by the front cover side cover plate **138** and the axially extending portion **148**, during installation and does not relax to a free state. The pulling bias spring **158** is configured to exert a continuous axial load onto the front side cover plate **138** and the axially extending portion **148** so as to pull the intermediate flange **144** and the front side cover plate **138** together, i.e., in opposite axial directions, which creates friction and may be beneficial NVH performance.

(21) The pulling bias spring **158** includes a body **160** contacting the front side cover plate **138**, and at least one tab **162** engaged with the axially extending portion **148**. The body **160** is arranged between the front cover **102** and the front side cover plate **138**. Specifically, the body **160** contacts an axial side of the front cover side cover plate **138** that faces the front cover **102**.

(22) The tab(s) **162** extend radially inwards from the body **160**, e.g., an inner diameter thereof, and extend axially from the body **160** towards the turbine assembly **106**. The tab(s) **162** of the pulling bias spring **158** engage(s) the axially extending portion **148** axially between the body **160** and the body **146**. That is, the axially extending portion **148** includes a retaining feature configured to retain the pulling bias spring **158** in an installed state. The retaining feature is disposed axially between the body **160** and the body **146** to maintain deflection of the tab(s) **162**. As shown in FIGS. **1-2**, the retaining feature may be at least one slot **164** configured to receive one respective tab **162**. The slot(s) **164** may be disposed axially between the end **150** and the body **146**. Specifically, the slot(s) **164** may be disposed between the body **146** and the axial side of the front cover side cover plate **138** that faces the front cover **102**. The slot(s) **164** may extend, at least partially, radially through the axially extending portion **148**. In an installed state, the tab(s) **162** are disposed in the corresponding slot(s) **164**. In the installed state, the tab(s) **162** exert a force on the axially extending portion **148** in the slot(s) **164** and on the front cover side cover plate **138** so as to pull the slot(s) **164** towards the axial side of the front cover side cover plate **138** that faces the front cover **102**. To preload the pulling bias spring **158** in this example, the body **160** is contacted against the front cover side cover plate **138**. Then, the tab(s) **162** are pushed axially against the end **150** of the axially extending portion **148** such that the tab(s) **162** deflect radially outwardly. The pulling bias spring **158** axially slides along the axially extending portion **148** until the tab(s) **162** reach the corresponding slot(s) **164** and deflect radially inwardly to engage the respective slot(s) **164**.

(23) As another example, as shown in FIGS. **3-6**, the retaining feature may be a flange **364** disposed at the end **150**. In the example shown in FIG. **3**, the flange **364** may extend continuously along the end **150**. The flange **364** extends radially outwardly from the end **150**. The flange **364** includes an axial side that faces the body **146**, which is disposed axially between the body **146** and the axial side of the front cover side cover plate **138** that faces the front cover **102**. In the installed state, the tab(s) **162** contact the flange **364**. In this situation, the tab(s) **162** exert a force on the axial side that faces the body **146** and on the front cover side cover plate **138** so as to pull the flange **364** towards the axial side of the front cover side cover plate **138** that faces the front cover **102**. To preload the pulling bias spring **158** in this example, the body **160** is contacted against the front cover side cover plate **138**. Then, the tab(s) **162** are pushed axially against the flange **364** of the axially extending portion **148** such that the tab(s) **162** deflect radially outwardly. The pulling bias spring **158** axially slides along the axially extending portion until the tab(s) **162** move axially past the flange **364**, which allows the tab(s) **162** to deflect radially inwardly and engage the axial side of the flange **364** facing the body **146**.

(24) In the example shown in FIGS. **5-6**, the flange **364** may extend partially along the end **150**. In this situation, the axially extending portion **148** may further include radial stops **466** circumferentially spaced from each other. The flange **364** is disposed circumferentially between the radial stops **466**. The radial stops **466** extend axially from the end **150** towards the body **146**. The radial stops **466** extend radially outwardly of an outer circumference of the axially extending portion **148**. The radial stops **466** are configured to radially constrain the pulling bias spring **158**, i.e., limit rotation of the pulling bias spring **158** such that the tab(s) **162** remain engaged with the

flange **364**. To preload the pulling bias spring **158** in this example, the body **160** is contacted against the front cover side cover plate **138** with the tab(s) **162** being circumferentially offset from the flange **364**. The body **160** is then rotated to align the tab(s) **162** with the flange **364**. In this situation, each tab **162** engages one radial stop **466** and is deflected radially inwardly. Once the tab(s) **162** clears the respective radial stops **466** in a circumferential direction, the tab(s) **162** deflect radially inwardly and engage the axial side of the flange **364** facing the body **146**. Further, the tab(s) **162** are arranged, at least partially, radially inside of the radial stops **466**, which prevents further rotation of the pulling bias spring **158**.

(25) By pulling the axially extending portion **148** towards the front cover side cover plate **138**, and specifically, an inner diameter thereof, the pulling bias spring **158** allows the body **146** of the intermediate flange **144** to directly contact the front cover side cover plate **138** radially outside of the arc springs **140**. That is, the pulling bias spring **158** is configured to pull the front side cover plate **138** and intermediate flange **144** into contact with each other without transferring axial loads to additional components in the damper assembly **108**. As such, a stack path for the pulling bias spring **158** is defined by a first axial offset between a portion of the front cover side cover plate **138** contacting the pulling bias spring **158** and a portion of the front cover side plate **138** contacting the intermediate flange **144** and a second axial offset between a portion of the intermediate flange **144** contacting the front cover side plate **138** and a portion of the intermediate flange **144** contacting the pulling bias spring **158**. The stack path for the pulling bias spring **158** is thereby reduced compared to a stack path for typical pushing bias springs, which further include axial offsets resulting between contacts of additional components in the damper assembly **108**.

(26) Embodiments according to the present disclosure provide various advantages including providing a bias spring that pulls the intermediate flange and the one cover plate together, which reduces a stack path for creating friction between the intermediate flange and the one cover plate. By reducing the stack path, a variation in install height of the diaphragm spring can be reduced as compared to bias springs that push components into contact with each other, which can result in more consistent hysteresis, loads, and stresses for the bias spring while satisfying packaging constraints of the torque converter.

(27) While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the disclosure that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, to the extent any embodiments are described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics, these embodiments are not outside the scope of the disclosure and can be desirable for particular applications.

LIST OF REFERENCE NUMBERS

(28) **100** torque converter **102** front cover **104** impeller assembly **106** turbine assembly **108** damper assembly **110** impeller shell **112** impeller blade **114** impeller hub **116** blade supporting portion **118** radially extending wall **120** turbine shell **122** turbine blade **124** blade supporting portion **126** outer radial extension **128** inner radial extension **130** axially extending inner circumferential section **132** hub **136** cover plate **138** cover plate **140** spring **142** drive flange **144** intermediate flange **146** body

148 axially extending portion **150** end **152** pushing bias spring **154** centering plate **156** bushing **158** pulling bias spring **160** body **162** tab **164** slot **364** flange **466** radial stops A central axis

Claims

1. A torque converter, comprising: a front cover arranged to receive torque; an impeller having an impeller shell non-rotatably connected to the front cover; a turbine in fluid communication with the impeller and including a turbine shell; a damper assembly disposed axially between the front cover and the turbine shell, the damper assembly including: a first cover plate non-rotatably connected to the turbine shell; a second cover plate non-rotatably connected to the first cover plate and disposed axially between the front cover and the first cover plate; an intermediate flange disposed axially between the first cover plate and the second cover plate; and a bias spring engaged with the intermediate flange and the second cover plate, the bias spring being configured to pull the second cover plate into contact with the intermediate flange.
2. The torque converter of claim 1, wherein the damper assembly further includes a drive flange disposed axially between the first and second cover plates, the drive flange being arranged to non-rotatably connect to a transmission input shaft.
3. The torque converter of claim 2, wherein the intermediate flange is disposed axially between the drive flange and the second cover plate, the intermediate flange being radially spaced from the transmission input shaft.
4. The torque converter of claim 2, further comprising a further bias spring engaged with the turbine shell and the first cover plate, the further bias spring being configured to push the first cover plate into contact with the drive flange.
5. The torque converter of claim 4, wherein the further bias spring is disposed radially inside of the intermediate flange.
6. The torque converter of claim 1, wherein the damper assembly further includes an arc spring supported by the first cover plate and the second cover plate, wherein the bias spring is disposed radially inside inward of the arc spring.
7. The torque converter of claim 6, wherein the intermediate flange and the second cover plate are configured to contact each other radially outside of the arc spring.
8. The torque converter of claim 1, wherein the intermediate flange includes an axially extending portion disposed radially inside inward of the second cover plate, the bias spring being engaged with the axially extending portion.
9. The torque converter of claim 8, wherein the axially extending portion includes a slot extending, at least partially, radially therethrough, the bias spring being disposed, at least partially, in the slot.
10. The torque converter of claim 8, wherein the axially extending portion includes a flange extending radially outwardly from the axially extending portion, the bias spring being in contact with the flange.
11. The torque converter of claim 10, wherein the axially extending portion includes two radial stops circumferentially spaced from each other and configured to radially constrain the bias spring, the flange being disposed circumferentially between the two radial stops.
12. The torque converter of claim 1, wherein the bias spring includes a body in contact with the second cover plate and a tab extending radially inwardly from the body, the body being disposed axially between the second cover plate and the front cover.
13. The torque converter of claim 12, wherein the intermediate flange includes a retaining feature disposed axially between the body and the first cover plate, the retaining feature being configured to deflect the tab relative to the body towards the first cover plate.
14. A damper assembly for a torque converter, comprising: a first cover plate arranged to receive torque; a second cover plate non-rotatably connected to the first cover plate and axially spaced from the first cover plate; an intermediate flange disposed axially between the first cover plate and

the second cover plate; and a bias spring engaged with the intermediate flange and the second cover plate, the bias spring being configured to pull the second cover plate into contact with the intermediate flange.

15. The damper assembly of claim 14, further comprising an arc spring supported by the first cover plate and the second cover plate, wherein the bias spring is disposed radially inward of the arc spring.

16. The damper assembly of claim 15, wherein the intermediate flange and the second cover plate are configured to contact each other radially outside of the arc spring.

17. The damper assembly of claim 14, wherein the bias spring includes a body in contact with the second cover plate and a tab extending radially inwardly from the body, the second cover plate being disposed, at least partially, between the body and the first cover plate.

18. The damper assembly of claim 17, wherein the intermediate flange includes a retaining feature disposed axially between the body and the first cover plate, the retaining feature being configured to deflect the tab relative to the body towards the first cover plate.

19. The damper assembly of claim 14, wherein the intermediate flange includes an axially extending portion disposed radially inward of the second cover plate, the bias spring being engaged with the axially extending portion.

20. The damper assembly of claim 19, wherein the axially extending portion includes a retaining feature configured to engage the bias spring, the retaining feature being one of a slot or a radially extending flange.
